

Program : Diploma in Robotic Process Automation	
Course Code : 4302	Course Title: Advanced Robotic Process Automation
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 45

Course Objectives:

- To understand the advanced concepts of robotic process automation.
- To Acquire knowledge of UiPath and its applications.
- To learn UiPath connection with database and the basics of blue prism
- To learn automation anywhere tool and Bot creation
- To understand debugging and exception handling of RPA tools.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge of Robotic process automation.		Introduction to Robotic Process Automation	3

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the RPA advanced concepts and the ability to differentiate it from other automation.	12	Understanding Apply

CO2	Acquire knowledge of UiPath and its applications	20	
CO3	Applying the advanced concepts of Ui Path and understanding the basic concepts of Blue prism.	14	Apply
CO4	Understand automation anywhere tool and bot creation	12	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3			2	3	3	
CO2	3				3	3	
CO3	3				3	3	2
CO4	3			3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the advanced concepts of Robotic Process Automation		
M1.01	RPA Solution Architecture Patterns: Key Considerations, Input Data Handling Solution Pattern	1	Understanding
M1.02	Exception Handling, Transaction Logging	1	Understanding
M1.03	Credential Management	1	Understanding
M1.04	Secure Execution, Monitoring and Reporting	1	Understanding

M1.05	Advanced Automation concepts and techniques: Text, Image & Advanced Citrix Automation	1	Understanding
M1.06	Introduction to Text & Image, Automation-Image based automation	1	Understanding
M1.07	Keyboard based automation, OCR and image based automation	1	Understanding
M1.08	Information Retrieval	1	Understanding
M1.09	Advanced Citrix Automation challenges, Guides/text activities, Guides/ocr-activities	2	Understanding
M1.10	Image-activities,Best Practices	1	Understanding
M1.11	Using tab for Images, Starting Apps	1	Understanding

Contents:

RPA Solution Architecture Patterns – Key Considerations, Input Data Handling Solution Pattern, Exception Handling, Transaction Logging, Credential Management, Secure Execution, Monitoring and Reporting

Advanced Automation concepts and techniques: Text, Image & Advanced Citrix Automation -Introduction to Text & Image, Automation-Image based automation, Keyboard based automation, OCR-and-image-automation, Information Retrieval, Advanced Citrix Automation challenges, Guides/text-activities, Guides/ocr-activities, Image-activities,Best Practices, Using tab for Images, Starting Apps.

CO2	Acquire knowledge of UiPath and its applications		
M2.01	UI automation concepts introduction –UiPath Building blocks	1	Understanding.
M2.02	Interfaces	1	Understanding
M2.03	Explain Debugging concepts	1	Understanding.
M2.04	Explain State Machines	1	Understanding.
M2.05	Describe the functions and differences between variables and arguments	1	Understanding.

M2.06	How arguments are used?	1	Understanding
M2.07	Describe the importance and reasons why data manipulation is used	1	Understanding
M2.08	Excel and Data Tables :Describe Microsoft Excel and data table functions, and how Excel activities are used for data manipulation	2	Apply
M2.09	Describe Data tables	1	Apply
M2.10	What is Data Scraping and Types?	1	Understanding
M2.11	What is web Scraping and Types?	1	Understanding
M2.12	Differentiate web-scraping and data Scraping with example	1	Understanding
M2.12	Mail and PDF :How is PDF-data extracted?	1	Apply
M2.13	M2.13 Describe the functions used to extract data from a .pdf file	1	Apply
M2.14	M2.14 Identify and explain how e-mail automation is used	1	Apply
M2.15	Exception and Debugging :Explain the importance of exception handling and how it can be implemented	2	Apply
M2.16	Explain Logging	1	Understanding
M2.17	Explain how debugging can be performed?	1	Understanding
M2.18	Explain Tracing ?	1	Understanding
	Series Test – I	1	

Contents:

UI automation concepts–UiPath Building blocks introduction- Interfaces, Debugging, State Machines, Managing-arguments, Using-arguments.

Excel and Data Tables : Data Manipulations, Excel Automation, Data tables. Scraping: web-scraping with example, data-scraping with example.

Mail and PDF : PDF-data-extraction, Email-automation.

Exception and Debugging: Exception Handling, Logging, Debugging, Tracing

CO3	Understand UiPath connection with database and the basics of blue prism		
M3.01	Database : Explain Connecting with Database	1	Understanding
M3.02	How can Execute Query with Database	2	Apply
M3.03	UiPath Orchestrator: Orchestrator Overview	1	Understanding.
M3.04	Explain Orchestrator Hosting	1	Understanding.
M3.05	Identify Robots Hosting	2	Apply
M3.06	Processing Hosting, Process Monitoring	2	Understanding
M3.07	Blueprism: Blue Prism Administration	1	Understanding
M3.08	Explain Users, Roles and Permission	1	Understanding
M3.09	Interface-Authentication in Blue Prism	1	Understanding
M3.10	Explain Studio-Business objects and processes	1	Understanding
M3.11	Define Process studio, Object studio	1	Understanding
Contents: Database : Connecting with Database, Executing Query with Database UiPath Orchestrator: Orchestrator Overview, Orchestrator Hosting, Robots Hosting, Processing Hosting, Process Monitoring Blueprism: Blue Prism Administration, Users, Roles and Permission, Interface Authentication in Blue Prism, Studio-Business objects and processes, Process studio, Object studio			
CO4	Understand automation anywhere tool and Bot creation		
M4.01	Automation anywhere: Introduction, Automation Anywhere Architecture	1	Understanding
M4.02	Web Control Room and Client- Introduction	1	Understanding
M4.03	Features Panel - Dashboard (Home, Bots, Devices, Audit, Workload, Insights)	1	Understanding

M4.04	Features Panel – Activity (View Tasks in Progress and Scheduled Tasks)	1	Understanding
M4.05	Bots (View Bots Uploaded and Credentials)	1	Understanding
M4.06	Devices (View Development and Runtime Clients and Device Pools)	1	Understanding
M4.07	Workload (Queues and SLA Calculator)	1	Understanding
M4.08	Audit Log (View Activities Logged which are associated with Web CR)	1	Understanding
M4.09	Administration (Configure Settings, Users, Roles, License and Migration)	1	Understanding
M4.10	Demo of Exposed API's – Conclusion – Client introduction and Conclusion	1	Understanding
M4.11	Bot Creator: Bot Creator overview, Bot Runner overview	1	Understanding
M4.12	M4.12 Deployed components, Bot design considerations	1	Understanding
M4.13	Series Test – II	1	
Contents: <i>Automation anywhere: Introduction, Automation Anywhere Architecture, Web Control Room and Client- Introduction- Features Panel - Dashboard (Home, Bots, Devices, Audit, Workload, Insights) , Features Panel – Activity (View Tasks in Progress and Scheduled Tasks) - Bots (View Bots Uploaded and Credentials) - Devices (View Development and Runtime Clients and Device Pools) - Workload (Queues and SLA Calculator) - Audit Log (View Activities Logged which are associated with Web CR) - Administration (Configure Settings, Users, Roles, License and Migration) - Demo of Exposed API's – Conclusion – Client introduction and Conclusion</i> <i>Bot Creator: Bot Creator overview, Bot Runner overview, Deployed components, Bot design considerations</i>			

Text / Reference

T/R	Book Title/Author
T1	Alok Mani Tripathi, “ Learning Robotic Process Automation ”, Packt Publishing, 2018

R1	Frank Casale , Rebecca Dilla, Heidi Jaynes , Lauren Livingston, “ Introduction to Robotic Process Automation: a Primer ”, Institute of Robotic Process Automation,1st Edition 2015.
R2	Richard Murdoch, Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant ”, Independently Published, 1st Edition 2018.
R3	Adeel Javed, Anum Sundrani, Nadia Malik, Sidney Madison Prescott,” Robotic Process Automation using UiPath StudioX A Citizen Developers Guide to Hyperautomation ” 2021 Edition
R4	Srikanth Merianda,” Robotic Process Autom Robotic Process Automation using UiPath StudioX A Citizen Developers Guide to Hyperautomation 2021 Editionation Tools, Process Automation and their benefits: Understanding RPA and Intelligent Automation ”, Consulting Opportunity Holdings LLC, 1st Edition 2018
R5	Lim Mei Ying, “ Robotic Process Automation with Blue Prism Quick Start Guide: Create software robots and automate business processes ”, Packt Publishing, 1st Edition 2018.
R6	Vaibhav Srivastava,” Getting started with RPA using Automation Anywhere ”

Online Resources

Sl.No	Website Link
1	https://www.w3schools.com/html/
2	https://www.w3schools.com/xml/
3	https://www.udemy.com/course/robotic-process-automation/
4	https://www.javatpoint.com/rpa
5	https://intellipaat.com/blog/tutorial/rpa-tutorial/
6	https://bpdocs.blueprism.com
7	https://www.uipath.com
8	https://docs.automationanywhere.com/bundle/enterprise

9	https://www.guru99.com/automation-anywhere-tutorial.html
10	https://www.guru99.com/blue-prism-tutorial.html
11	https://www.tutorialspoint.com/blue_prism/blue_prism_introduction_to_rpa.htm